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Understanding accounting students' intentions to use digital badges to showcase employability skills

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ABSTRACT

By incorporating employability skills within the accounting curriculum, universities face the issue of how best to recognise students' achievements. Digital badges are emerging as a potential means to recognise such achievements. Being shareable on social media sites, such as LinkedIn, and on other platforms, badges allow students to showcase their attainment of employability skills to potential employers. As student acceptance is a prerequisite to the success of this technology, this exploratory study examines accounting students' perceptions of badge usefulness and ease of use, and whether these influence their intentions to use them for job applications. We also examine factors that contribute to these perceptions. A survey is undertaken of accounting students within a single university. Results suggest that both subjective norm and perceived usefulness directly influence intentions, while job application relevance and perceptions of external control have important indirect effects. Implications for universities are discussed and future research opportunities identified.

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Employability skills; digital badges; perception of badge usefulness; showcase competencies; student intention

Introduction

The aim of this study is to explore students' intentions to use university-issued digital badges for the purposes of showcasing employability skills and to identify factors that influence these intentions. The application of digital badge technology in higher education is a rapidly developing and nascent practice. A digital badge is a form of digital credential which typically consists of a clickable graphical icon containing verifiable and secure metadata concerning the recipient's achievement(s) or skill(s) for which the badge was awarded, and details such as the name of the issuer, date of issuance, and the criteria for which the badge was awarded (Carey & Stefaniak, 2018; Dyjur & Lindstrom, 2017; Sousa-Vieira et al., 2022).

In recent years, the tertiary sector has faced considerable pressure from both industry and government to address perceived shortfalls in graduates' employability skills

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(Kavanagh & Drennan, 2008; Mahon, 2022; Tan et al., 2022). As Sin and McGuigan (2013, p. 524) observe, universities are increasingly being 'held accountable for the development of graduates who are both knowledgeable in their respective fields of study and ready for their working career'. Universities have responded by developing and implementing graduate attributes across and within academic programmes (Kavanagh & Drennan, 2008; Oliver & Jorre de St Jorre, 2018). Graduate attributes are 'the qualities, skills and understandings a university community agrees its students should develop ... [and] include, but go beyond, ... disciplinary expertise or technical knowledge' (Bowden, Hart, King, Trigwell & Watts, 2000). They encompass a wide range of generic, transferable, soft skills, such as teamwork, problem-solving, and communication, and typically have a significant focus on graduate employability (Oliver, 2011, p. 2015).

However, the achievements of students with respect to employability skills are generally not captured by traditional academic transcripts (Jorre de St Jorre, 2020; Miller et al., 2020). As a result, students must find other means of evidencing and communicating their employability skills to differentiate themselves in the highly competitive graduate job market. Digital badges are a potential solution to this issue. Digital badges can communicate student's employability skills in a more flexible, timely and granular fashion than traditional academic transcripts. They may incorporate detail on assessment criteria and even examples of assessment submitted for assessment (Jorre de St Jorre, 2020). Furthermore, recipients can choose a variety of ways with which to display their badges, including sharing them on social media sites such as LinkedIn, Facebook or Twitter, or embedding them into personal webpages or email signatures (Dyjur & Lindstrom, 2017; Gibson et al., 2015; Shields & Chugh, 2017). Social media are increasingly being used by employers in the recruitment process to identify and vet potential candidates (Castrillon, 2021; Jorgensen, 2023). The potential of digital badges is particularly germane to accounting, where accounting employers, such as the Big 4, are increasingly placing emphasis on employability skills over academic performance (Agnew, 2016; Jackson et al., 2021).

Given the emerging nature of digital badges in higher education and the paucity of related research, our study is exploratory in nature. Based on the Technology Acceptance Model (TAM) literature (e.g. Davis et al., 1989; Davis, 1989; Venkatesh, 2000; Venkatesh & Bala, 2008; Venkatesh & Davis, 2000) we develop and apply a model of the general determinants of perceived usefulness, perceived ease of use and behavioural intention to adopt digital badges for job application purposes. The model is tested with survey data obtained from second and third year accounting students at a New Zealand university. Results suggest that accounting students would adopt digital badges to facilitate their job application activities. Furthermore, both subjective norm and perceived usefulness directly influence students' intentions to use digital badges, while job application relevance and perceptions of external control have important indirect effects. Additional analyses of 13 technology-specific issues identified from the literature highlighted further issues for future studies to consider.

Contribution

The research makes several contributions to practice and theory. First, the findings are directly relevant to higher education providers that may be considering supplementing

academic transcripts with digital badges that recognise students' employability skills. While investment in digital badge technology may bring several benefits, including raising awareness of the relevance of graduate attributes among students, enhancing universities' perceived value to external stakeholders, reputational effects due to the symbolic use of technology, and accreditation utility, the investment is not without risk. New innovations, such as digital badges, require user acceptance as a prerequisite for success. Knowledge of factors identified in this study as contributing to students' intentions to accept and voluntarily use digital badges as a means of signalling employability skills will enable higher education institutions to undertake targeted activities designed to increase the likelihood of their investments in badge programmes ultimately being successful.

Second, this study makes an early contribution to research on digital credentials in the accounting literature. As an exploratory study, it focuses on identifying the salient issues and laying a foundation for further in-depth studies.

Third, it adds to the accounting graduate employability skills literature in accounting. This literature has largely focused on identifying the generic skills expected of accounting graduates (Chaplin, 2017; Hayes et al., 2022; Kavanagh & Drennan, 2008; Low et al., 2016) and determining how these can be most effectively implemented in the accounting curriculum (Sin & McGuigan, 2013; Tan et al., 2022). This study contributes to this literature by drawing attention to the role that digital badges may play in enhancing students' awareness of the linkages between graduate attributes and future employment. This is important as many students find it difficult to articulate the general attributes they have obtained beyond disciplinary knowledge (Hill & Walkington, 2016).

Last, the study extends TAM to digital badges. TAM has been used extensively in the education research and is the most widely used theoretical model in the e-learning acceptance literature (Abdullah & Ward, 2016). In particular, the study demonstrates the value of specifying external factors as antecedents of perceived usefulness, ease of use, and behavioural intentions, consistent with Venkatesh and Bala (2008).

The next section discusses the literature relevant to employability skills and digital badges. This is followed by the study's research model and hypotheses. Next the study's research method is outlined, followed by the results and discussion. Last, we conclude with a summary of the research findings.

Literature review

Graduate attributes and employability skills

There is evidence of growing tension surrounding universities' purpose and accountability (Craig & Amernic, 2002; Menz, 2020; Star & Hammer, 2008) and scepticism about their value as they struggle 'to provide students with the necessary skills to succeed after graduation' (Menz, 2020, p. 115). It is claimed that universities are not doing enough to enable students to meet employers' expectations and needs (Menz, 2020). Employers are increasingly concerned about receiving 'work-ready' graduates from universities – that is, graduates that not only possess discipline-specific technical knowledge but also the necessary generic skills that enable them to competently perform in a work context (Hayes et al., 2022; Singhal, 2017). The accounting discipline

has echoed these concerns over several decades with questions raised over the work-readiness of its graduates (American Accounting Association, 1986; Accounting Education Change Commission, 1990; Albrecht & Sack, 2000; Kavanagh & Drennan, 2008; Bui & Porter, 2010).

One way in which universities have responded has been through the formal articulation of institutional and/or programme level graduate attributes. Graduate attributes encompass both employability skills (Sin & McGuigan, 2013; Tempone et al., 2012) and attributes relating to citizenship (Bowden et al., 2000). However, in practice, the concern with developing employability skills has tended to dominate over citizenship-focused attributes (Kensington-Miller et al., 2018). Unsurprisingly then, many studies have treated graduate attributes as synonymous with employability skills (Kensington-Miller et al., 2018).¹

Employability skills can be defined as ‘transferable core skill groups that represent essential functional and enabling knowledge, skills, and attitudes required by the twenty-first century workplace. They are necessary for career success at all levels of employment and for all levels of education’ (Overtoom, 2000, p. 2).² Perceived deficiencies in the employability skills of accounting graduates has led to extensive efforts by both the research community and professional bodies to identify the desirable skills of entry-level accounting professionals (for example, Birkett, 1993; Chaplin, 2017; Hayes et al., 2022; Kavanagh & Drennan, 2008; Low et al., 2016). Furthermore, such concerns have prompted professional accounting bodies internationally to require higher education institutions to take a measure of responsibility for employability skills (professional skills) development via institutional accreditation requirements. For example, both CPA Australia and Chartered Accountants Australia New Zealand (CAANZ) require accounting degree programmes of accredited academic institutions to cover a range of broad professional competency areas, including intellectual skills (e.g. critical thinking), interpersonal and communication skills (e.g. teamwork and negotiation skills), personal skills (e.g. attitudes and behaviours of a professional, such as time management and awareness of biases), and ethical principles, professional values and integrity.

Next, we consider the issue of the visibility of employability skills-related graduate attributes.

Making employability skills visible

The academic transcript is the principal record of a student’s academic achievements. It is traditionally limited to the reporting of such information as programme of study, GPA, courses (units/modules), and corresponding grades, but provides ‘little or no information about the context, criteria and standards associated with assessment’ (Jorre de St Jorre, 2020, p. 278). Importantly, it also ‘rarely capture[s] transferable skills required in the workplace, such as teamwork, communication and critical thinking’ (Jorre de St Jorre, 2020, p. 278), even when these are part of the formal graduate attributes of the institution. This makes them effectively ‘invisible to students and employers’ (Kensington-Miller et al., 2018, p. 1440). Moreover, these are the skills that ‘many employers are most interested in for the purpose of distinguishing between similarly qualified candidates for a post’ (Maina et al., 2021, p. 63).

Acknowledging these concerns, some universities have started to provide supplementary information via extended transcripts or co-curricular records (Hope, 2016). These may provide details of certain co-curricular or extra-curricular activities undertaken by students, such as clubs they have led or internships undertaken, that have been validated by the issuing institution and, in some cases, linked to corresponding graduate attributes. However, as Jorre de St Jorre (2020, p. 280) notes, these emerging documents ‘still do little to identify the criteria or standards of achievement that were met to achieve the degree’. Furthermore, and importantly, they are generally not provided in a format that is timely, secure, self-verifying, and useful for sharing on social media in a way that effectively facilitates the showcasing of students’ employability skills.

As the discussion that follows will suggest, digital badges can offer significant benefits over both traditional and emerging extended transcripts and co-curricular records. They are an efficient, portable, timely and self-validating means of showcasing students’ employability skills. Early successful applications in the area of graduate attributes have seen digital badges awarded on completion of generic skills-focused modules (Gogel et al., 2020) and on the basis of student submitted evidence-based portfolios (Miller et al., 2020).

What are digital badges?

A digital badge is an electronic signifier of an achievement, analogous to physical Boy Scout badges earned through the achievement of certain competencies. Digital badges are a secure, flexible and efficient way for educators, community groups and other professional organisations, to exhibit and reward participants for skills obtained in professional development or formal and informal learning.

Digital badges can be distinguished from the related terms, *micro-credentials* and *open badges*, which are sometimes mistakenly used interchangeably with the former. A *micro-credential* ‘is the record of the learning outcomes that a learner has acquired following a small volume of learning’ (European Commission, 2021, p. 1) whereas *digital badges* ‘are visual tokens used to acknowledge learning and achievement’ (Hartnett, 2021, p. 104). Digital badges may be applied to micro-credentials or, indeed, other forms of certification. *Open badges*, on the other hand, are a subset of digital badges and were introduced by the Mozilla Foundation in 2012. They are based on an open technical specification, Open Badges Infrastructure, which facilitates their public sharing (Tomić et al., 2017). Open badge recipients can ‘combine badges from different issuers and organise the collected badges ... in ways they find most suitable for telling a story of their skills and achievements’ (Tomić et al., 2017, p. 520).

2014 Digital badges consist of an image or graphical icon that links directly to verified information of learning experiences obtained from the awarding body. They are embedded with metadata, containing information about the recipient, what they had to do to obtain the badge, the name of the issuing institution, the date issued, the criteria for earning the badge, and sometimes links to the assessment that the badge recipient completed as part of receiving the badge (Bowen & Thomas, 2014). If applicable, they can also include an expiration date and/or details of revocation. Hence, digital badges provide more detailed information about what the recipient learned than a traditional paper certificate. By using metadata related to accomplishments, digital badges offer

opportunities to recognise non-formal, informal and professional learning accomplishments (Hartnett, 2021).

Educational institutions and other organisations that award digital badges typically contract with a third-party credentialing platform, such as Credly and Accredible, which issues and manages the digital badges that have been designed by the awarding body. Upon being issued, recipients typically have the option, via the credentialing platform, of sharing their badge on social media sites, such as LinkedIn, Facebook, and Twitter, emailing the badge, sharing a public link, or embedding the underlying code into an email signature or other HTML document, such as a personal website. Many platforms also permit digital badges to be secured on the blockchain, providing additional assurance that a badge has not been altered, faked or spoofed. When a viewer clicks on a badge, they will typically be taken to the credentialing platform's website where details of the badge's metadata will be displayed in a user-friendly format. If the badge is on the blockchain, they will also be able to further verify the credential.

The potential advantages of digital badges are many. First, they make micro-skills or credentials both visible and portable in an online environment (Shields & Chugh, 2017). Because they display a plethora of information with granularity, they also make the represented credentials more transparent and evidence-based, particularly if they are linked to assessment evidence (Jorre de St Jorre, 2020; Miller et al., 2020). The personalised and rich evidence of achievement also potentially contribute to digital badges being a more effective tool for differentiating between students than traditional transcripts (Miller et al., 2020). Moreover, they allow learners to reflect upon and track their own soft skills (Parker, 2015); document their professional learning development attained through stand-alone workshops and seminars (Dyjur & Lindstrom, 2017); and select and sequence the informal (non-credit) learning opportunities they want to create their own personal learning pathway (Gamrat et al., 2014).

Further, since the information on a digital badge is stored in a self-verifying machine-readable form, it is a much easier mechanism to search and verify that formal qualifications and skills are accurate (Shields & Chugh, 2017). Last, from an institutional perspective, digital badges seem a viable 'strategy for meeting accreditation requirements, particularly in areas that are historically challenging to instruct, assess, and report: for example, life-long learning and professionalism' (Gogel et al., 2020, p. 3).

Use of digital badges in higher education

Hartnett (2021, p. 104) argues that digital badge technologies 'are fundamentally changing the accessibility, organisation and credentialing of education'. They are useful to higher education institutions as they have potential to strengthen traditional degree programs by supporting complementary competency-based programs (Carey & Stefaniak, 2018; Sousa-Vieira et al., 2022). A large amount of learning that might otherwise go unrecognised through formal academic accreditation processes occurs outside of traditional classrooms (Wilson et al., 2016) and may be amenable to digital badges. The University of Central Oklahoma, for instance, has identified soft skills they expect graduates to demonstrate and then use badges to document and track these skills in addition to grades (Parker, 2015).

Badges also offer alternatives to traditional academic transcripts and can augment current formal accreditation structures (Hartnett, 2021). They enable viewers to verify issuer details, and determine the purpose for which it was issued, what the participant had to do in order to receive the badge, the evaluation criteria, the credential or skill obtained, and supporting evidence such as the actual assessed work (Parker, 2015; Shields & Chugh, 2017; Sousa-Vieira et al., 2022).

Digital badges also enable learners to showcase to potential employers skills and achievements that have been acquired outside the traditional curriculum, including accomplishments acquired through informal learning, at a more granular level than information captured in traditional courses or degrees (Dyjur & Lindstrom, 2017; Gamrat et al., 2014; Gibson et al., 2015; Goligoski, 2012; Sousa-Vieira et al., 2022). Further, higher education institutions may use digital badges as incentives to push students toward pursuing well defined goals and skills. They may be used to motivate engagement, participation, and achievement (Sousa-Vieira et al., 2022).

The development of open badges, however, has raised questions about how higher educational institutions will respond if a global badge ecosystem becomes an alternative, and perhaps dominant, credentialing system, and how they will adapt if companies begin hiring people on the basis of their LinkedIn badge collection rather than their degrees (Carey & Stefaniak, 2018).

In summary, digital badges may be used to set goals, motivate learners, encourage participation, represent achievements, communicate success in many contexts, and to provide status and recognition in an online community (Carey & Stefaniak, 2018; Dyjur & Lindstrom, 2017; Hartnett, 2021). Whilst acknowledging the varied roles of digital badges, the current study only focuses on one of these, namely their use in showcasing employability skills.

Issues and challenges with digital badges in higher education

As with most technology solutions, digital badges are not without their risks and challenges. Not all digital badges are created equal – some may have more value and credibility than others, depending on the issuer, the integrity of the assessment task(s) and related judgements, and the nature of any supporting evidence (Jorre de St Jorre, 2020; Randall & West, 2022). Furthermore, as the value of badges is tied to their perceived utility, badges for employability must be able to satisfactorily differentiate job candidates, and ‘represent what students are able to do at the completion of their course ... and reflect skills and knowledge valued by employers ...’ (Jorre de St Jorre, 2020, p. 284).

Owing to the ease with which virtually any organisation can create and award digital badges, their value may also be undermined by the variability in quality of badges across institutions (Abramovich, Schunn & Higashi 2013; Jorre de St Jorre, 2020). Furthermore, as badges are applied to a wide range of purposes, including recognising achievement of relatively low-level tasks and competencies, and for participation rather than achievement, there is a very real risk that the meaning of badges may be devalued by association (Carey & Stefaniak, 2018; Jorre de St Jorre, 2020).

Badge credibility is also vulnerable to the risk that they may be used inappropriately, for example, being forged or otherwise manipulated. According to Coleman (2018, p. 216), this risk ‘necessitates a means of encryption and authentication’. As the verifying

information associated with digital badges is hosted on the credentialing platform's website, much store must be placed in the quality of their security procedures, including whether blockchain technology is used. A somewhat related concern is the privacy of badge recipients' personal information held by the credentialing platform (Pitt et al., 2019). Again, badge recipients must rely on steps taken by the platform to secure personal information from unauthorised access and disclosure. Privacy issues also arise when badge recipients attempt to share their badges on social media websites, like LinkedIn, as the credentialing platform will require the recipient to grant ongoing access to their social media account to ensure that the badge's currency can be directly maintained by the issuer. Lastly, a likely concern for both students and universities is the long term continuity of service and existence of the credentialing platform.

Justification for the current study

As discussed earlier, traditional academic transcripts rarely capture employability skills making them invisible to both students and employers, while emerging forms of transcripts, such as extended transcripts and co-curricular records, provide limited information and are generally not provided in a format that is timely, secure and useful for sharing on social media in a manner that effectively facilitates the showcasing of students' employability skills.

These issues are important as labour market research suggests that the information available to both job seekers and employers is instrumental in determining the efficiency of the job matching process (Jovanovic, 1979; Bassi & Nansamba, 2022). Graduates typically have limited work experience and need a means of signalling difficult to observe employability skills to differentiate themselves from similarly academically qualified graduates. In the absence of credible signals on such skills, employers also find it difficult to differentiate between applicants and must rely on costly and potentially less effective internal selection procedures. Recent research suggests that trustworthy information on difficult to observe skills has positive labour market outcomes (Abebe, Carla, Fafchamps, Falco, Franklin & Quinn, 2021; Bassi & Nansamba, 2022), especially when that information can be easily and credibly shared (Carranza, Garlik, Orkin & Rankin, 2022).

The informativeness, portability, timeliness, and self-validating nature of digital badges suggest that they have a relative advantage as a credible means of showcasing students' employability skills relative to traditional academic transcript and emerging forms of extended and co-curricular transcripts whether used for the purposes of passive or active job searching. However, the showcasing benefit of digital badges will not be realised unless students are willing to embrace them. Therefore, it is critical to determine (1) whether students do perceive such badges to be useful, and if so, (2) the factors that influence their intentions to use them with a view to enhancing their future employment prospects.

The current study investigates these issues by examining students' perceptions about aspects of digital badges. Accordingly, the following research questions were developed:

RQ1: What are accounting students' perceptions of the usefulness and ease of use of digital badges that are designed to showcase their employability skills?

RQ2: What *general* factors influence accounting students' perceptions of the usefulness and ease of use of digital badges that are designed to showcase their employability skills?

RQ3: To what extent do accounting students intend to use digital badges to showcase their employability skills?

RQ4: What *specific* concerns do accounting students have about digital badges?

The first three questions will be addressed by the development and testing of a TAM-based research model, while the last question is considered in light of students' perceptions about the importance of specific issues noted in the literature.

The research model, hypotheses and underpinning theory are discussed next.

Research model and hypotheses

Technology acceptance model

Among information technology adoption studies, a vast number have favoured the Technology Acceptance Model (TAM) first developed by Davis in 1986 (Abdullah et al., 2016; Davis, 1989). TAM focuses on the attitudinal explanations of individual intentions to use a specific technology or service and typically explains around 40% of the variance in usage intentions and behaviour (Venkatesh & Davis, 2000). Abdullah et al. (2016) assert that '... TAM is a valid model that represents an important theoretical framework to explain and predict technology acceptance behaviour' (p. 77). Horst et al. (2007) argue this is because the model's roots lie in the successful theoretical frameworks as the theory of reasoned action (TRA) and its successor, the theory of planned behaviour (TPB). TAM and TPB are frequently used to explain end-users' adoption and acceptance of different kinds of IT-systems and applications (Horst et al., 2007). TAM focuses on perceived benefits and postulates that actual use and behaviour are dependent on the behavioural intention (BI).

Because of its generality and credibility, TAM has been extended and used extensively in education to explain users' intentions to use different types of e-learning and e-portfolio technologies. Indeed, Sumak, Hericko and Pusnik (2011) showed it being employed in 86% of e-learning acceptance studies. In a recent literature review of TAM in educational contexts, Granić and Marangunić (2019) conclude that TAM (and its variants) represents a credible model for evaluating a diverse range of educational technologies.

A common criticism of the basic TAM model, however, is that it is too generic and fails to provide 'actional guidance' to management (Venkatesh & Bala, 2008, p. 274). As a result, researchers have developed extended and context-specific TAM models which attempt to identify *general* determinants of perceived usefulness (PU) and perceived ease of use (PEOU) (Venkatesh & Bala, 2008; Abdullah et al., 2016). In the current study, we develop an extended and context-specific TAM based on Venkatesh & Bala's TAM3 in order to understand (1) accounting students' intentions to adopt digital badges for the purposes of showcasing their employability skills, (2) their perceptions of both badge usefulness and ease of use for that purpose, and (3) the relative importance of several general determinants. We now discuss the hypotheses underlying the study's research model (depicted in Figure 1).³

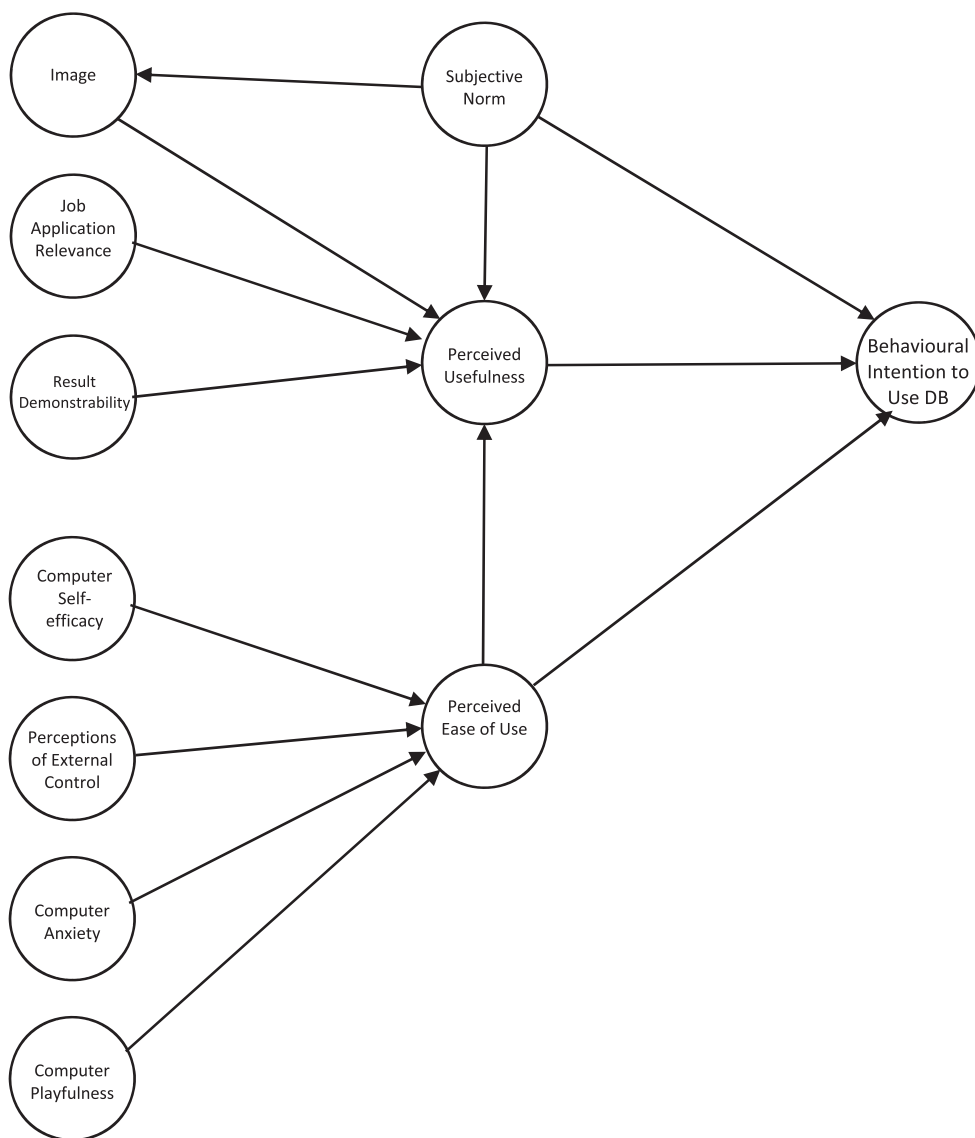


Figure 1. Caption: Research model.

Hypothesis development

In a recent meta-analysis study, the most important constructs in TAM are confirmed to be *perceived usefulness* and *perceived ease of use* (Chen et al., 2015). This is consistent with Davis et al.'s (1989) view that users' acceptance or rejection of technologies is mainly influenced by these two constructs. Perceived usefulness refers to 'the degree to which a person believes that using a particular system would enhance his or her job performance' while perceived ease of use is defined as 'the degree to which a person believes that using a particular system would be free of effort' (Davis, 1989). Perceived ease of use is also theorised to influence perceived usefulness, as '... the easier the system is to use, the more useful it can be' (Venkatesh & Davis, 2000, p. 187).

H1: Perceived usefulness has a positive direct effect on students' intention to use digital badges in applying for jobs.

H2: Perceived ease of use has a positive direct effect on students' intention to use digital badges in applying for jobs.

H3: Perceived ease of use has a positive direct effect on students' perceived usefulness of digital badges in applying for jobs.

Adding general determinants (external variables) to the basic TAM not only enhances researchers ability to explain adoption behaviour but also facilitates the identification of reasons why the technology may not be adopted, thereby enabling researchers and practitioners to develop corrective responses (Abdullah et al., 2016). Four variables from Venkatesh and Bala's (2008) study were added to the model to represent the effects of social influence (*subjective norm* and *image*) and cognitive instrumental processes (*job application relevance* and *results demonstrability*).

The first of these, *subjective norm*, is 'the degree to which an individual perceives that most people who are important to him think he should or should not use the system' (Venkatesh & Bala, 2008, p. 277). In the context of the current study, it corresponds to a student's perceived expectations of members in his or her environment (e.g. peers, parents/guardians, mentors, university, etc.) that they ought to use digital badges in the job application process. Generally, Venkatesh and Bala (2008) believe that the effects of external factors on behavioural intention will be mediated by perceived usefulness or perceived ease of use. However, consistent with Venkatesh and Davis (2000), they acknowledge that there could also be direct effect of subjective norm on behavioural intention when a person chooses to adopt a technology despite them not necessarily being favourable towards it (or its consequence), in situations where 'they believe one or more important referents think they should, and they are sufficiently motivated to comply' (Venkatesh & Davis, 2000, p. 187). While neither Venkatesh and Davis (2000) nor Venkatesh and Bala (2008) found a significant direct effect of subjective norm on intention to use, several earlier and later studies have (Taylor & Todd, 1995; Ursavaş et al., 2019). Further, a meta-analysis by Schepers and Wetzels (2007) confirmed a significant influence of subjective norm on behavioural intention.

H4: Subjective norm has a positive direct effect on students' intention to use digital badges in applying for jobs.

Subjective norm may also influence behavioural intention indirectly via perceived usefulness through an internalisation process, whereby an individual who perceives that someone important to them thinks they should adopt a technology, incorporates that person's belief into their own belief structure (Venkatesh & Davis, 2000).

H5: Subjective norm has a positive direct effect on students' perceived usefulness of digital badges in applying for jobs.

Image is 'the degree to which use of an innovation is perceived to enhance one's ... status in one's social system' (Moore & Benbasat, 1991, p. 195). Subjective norm may directly impact image when important referents believe a student should adopt technology like digital badges, and the student perceives that complying with that belief will likely raise their standing within the group (Venkatesh & Davis, 2000). Further, students

may perceive indirect benefits from badge use due to image enhancement per se, over and above any usage benefits directly attributable to the adoption of digital badges (Venkatesh & Davis, 2000).

H6: Subjective norm has a positive direct effect on image.

H7: Image has a positive direct effect on perceived usefulness of digital badges in applying for jobs.

Venkatesh and Davis (2000, p. 190) argue that people form judgements about the usefulness of technology based on cognitive comparisons between ‘what a system is capable of doing with what they need to get done.’ Accordingly, we include two corresponding constructs from Venkatesh and Bala (2008): *job relevance* [‘The degree to which an individual believes that the target system is applicable to his or her job’ (p.277)] and *result demonstrability* [‘The degree to which an individual believes that the results of using a system are tangible, observable, and communicable’ (p.277)]. Students will have a sense of what tasks are required in applying for jobs and on that basis would be expected to make assessments of the compatibility and consistency of digital badges with respect to those tasks. Furthermore, students will be more favourably inclined towards digital badges to the extent that they can readily discern a link between their use and successful outcomes from their use.

H8: Job application relevance has a positive direct effect on perceived usefulness of digital badges in applying for jobs.

H9: Result demonstrability has a positive direct effect on perceived usefulness of digital badges in applying for jobs.

We now focus on the remaining four external constructs, which are all posited to affect students’ perceptions of the ease of using digital badges. Based on anchor and adjustment framing, Venkatesh (2000) suggests that technology users will form *early* perceptions of perceived ease of use based on several anchors associated with individuals’ general beliefs regarding computers and their use. The four constructs are: *computer self-efficacy*, *perceptions of external control*, *computer anxiety*, and *computer playfulness*.

Computer self-efficacy relates to an individual’s beliefs about their personal ability to effectively use computers. In an e-learning context, Abdullah and Ward’s (2016) meta-analysis confirmed a positive association between this construct and perceived ease of use across 41 studies, while both Abdullah et al. (2016) and Chen et al. (2012) found a positive association with respect to e-portfolio adoption. The ways in which recipients can use their digital badges vary in sophistication, ranging from simply granting access (via the credentialing platform) to specific social media sites in order to display a badge on the recipient’s profile page, to embedding code relating to the badge into an HTML-based document. As such, it would be reasonable to expect that individual badge use will depend, in part, on the recipient’s perceptions of their own computer abilities. *Perceptions of external control* relate to an individual’s beliefs about the availability of necessary organisational and technical resources and support structures to facilitate their effective use of a technology. University support of technology adoption and use can be a significant enabling force (Teo, 2010; Unal & Uzun, 2021). Universities can support digital badge adoption in a variety of ways, including the provision of hands-

on training, online tutorials, regular communications to students, online FAQs, help desks, etc. *Computer anxiety*, unlike the previous two constructs, is expected to negatively affect perceived ease of use, and reflects ‘an individual’s apprehension, or even fear, when faced with the possibility of using computers’ (Venkatesh et al., 2003, p. 349). Computer anxiety is also a predictor of privacy attitudes (Gerber et al., 2018), and privacy is a ubiquitous concern among users of digital badges (Pitt et al., 2019). Abdullah and Ward (2016), reviewed e-learning adoption studies and found that 59% indicated a negative relationship between computer anxiety and perceived ease of use, with an average effect size of -0.199 . Last, *computer playfulness*, an intrinsic motivation factor, reflects a user’s degree cognitive spontaneity in interacting with computers (Venkatesh & Bala, 2008). It is expected to be positively associated with perceived ease of use. Padilla-Meléndez et al. (2013) reviewed 10 TAM studies and found that computer playfulness was positively associated with perceived ease of use in all but one case. Depending on where they are to be displayed, users of digital badges have discretion over their grouping, sizing, and positioning. This is particularly the case with badges embedded into HTML documents. There also numerous potential outlets for badges. Consequently, greater experimentation with such parameters might be expected from users that rate highly on computer playfulness.

H10: Computer self-efficacy has a positive direct effect on perceived ease of use of digital badges in applying for jobs.

H11: Perceptions of external control has a positive direct effect on perceived ease of use of digital badges in applying for jobs.

H12: Computer anxiety has a negative direct effect on perceived ease of use of digital badges in applying for jobs.

H13: Computer playfulness has a positive direct effect on perceived ease of use of digital badges in applying for jobs.

Research method

Survey instrument

Data for the study was collected using an online survey instrument. The survey items are presented in Appendix 1 and correspond to the research model’s 11 constructs (see Figure 1). Three to four reflective items were used to measure each of the model’s constructs. Where necessary, the items were adapted from those used in the TAM literature in order to accommodate the specific technology (digital badges) and focal purpose (signalling job-relevant skills and competencies). Karahanna and Straub (1999) argue that better explanatory power can be achieved when measures are specific to the target technology. TAM items have been similarly adapted in other studies in the e-learning and e-portfolio literatures (for instance, Abdullah et al., 2016). An additional 13 items (see Table 5) were included in the research instrument to address specific issues not deemed to be captured by the ‘general’ TAM items. These latter items were identified based on the review of the literature and were used for additional analyses.

The level of knowledge of digital badges among accounting students was expected to be limited. Consequently, one of the researchers presented an introduction to the

technology (and its applications) in class and through the inclusion of an informational video embedded within the online survey instrument along with additional narrative information. This material had to be viewed and read before the research participant could advance to the survey questions. Among other things, the video and narrative over-viewed the technology underlying digital badges, the nature of third-party digital credential organisations (like Credly), and discussed potential applications of badges in university and professional accounting settings (including their use in showcasing achievement of competencies and graduate attributes).⁴

Sample size determination

As will be discussed further in the Results section, we used the Partial Least Squares (PLS) method to structural equation modelling. PLS is noted for ‘generally achieving high levels of statistical power with small sample sizes’ (Hair et al., 2017, p. 19). The small sample size advantage of PLS has wide support in the literature (e.g. Barclay et al., 1995; Chin, 1998; Chin et al., 2003; Gefen et al., 2011). A common rule of thumb for determining the minimum sample size when using PLS is the ‘10 times’ rule. This suggests that for a reflective model, the sample size should be at least 10 times the largest number of structural paths directed at a construct anywhere in a path model (Goodhue et al., 2012; Hair et al., 2017). In respect of the current study, PU has the most incoming paths (5), suggesting a minimum sample size of 50. However, Hair et al. suggest that a more appropriate approach is to determine the sample size ‘by means of power analyses based on the part of the model with the largest number of predictors’ (p. 25). Based on Cohen (1992), Hair et al. (2017) provide a table for such a purpose. For a desired significance level of 5%, a maximum number of arrows pointing to a construct of 5, and a minimum R^2 of 0.25, a minimum sample of 45 is suggested. If the minimum R^2 is 0.50, then a sample of at least 20 would be warranted. In this study, we erred on the side of caution and aimed for a sample exceeding 50.

Administration of the survey

The research was undertaken within the Business School of a New Zealand university.⁵ Near the end of the academic year an invitation was posted on the online news forums of four accounting-related courses seeking participants to complete an anonymous and

Table 1. Demographics of surveyed students.

	Frequency	Percentage
Gender		
Male	17	29.8
Female	40	70.2
Age		
24 or under	48	84.2
25–34	8	14.0
35–44	1	1.8
45–54	0	0.0
55 or above	0	0.0

confidential online survey.⁶ Students in these courses were at the end of their second or third (and final) year of study. A total of 57 usable responses were received and the gender and age demographics are presented in Table 1. Of the responses, 70.2 percent were female accounting students, while 84.2 percent of participants were 24 years old or younger. Testing of earlier versus later responses did not reveal any systematic differences in responses – significant mean differences were only found on two indicator items.

An additional question asked participants to indicate their level of familiarity with digital badges on a five-point scale with ‘Not at all familiar’ (1) and ‘Very familiar’ (5) end-points. The mean response was 2.39, indicating that participants generally had a modest awareness of digital badges. This result was not unexpected, given that the university in which the research was conducted did not yet issue digital badges to students.

Results and discussion

The composite-based PLS approach to structural equation modelling was used in this study. Specifically, SmartPLS 4 (version 4.0.8.4) was used. Unlike covariance-based structural equation modelling, PLS ‘makes practically no assumptions about the underlying data’ (Hair et al., 2017, p. 18). While PLS may use reflective or formative measurement models, in this study and consistent with the prior literature, indicators for all constructs were treated as reflective. PLS uses a nonparametric bootstrap procedure for determining the significance of outer loadings, and path coefficients. Following Hair et al.’s (2017) recommendation, 5,000 bootstrap samples was set as the default for the analysis.

Measurement model

Table 2 presents the statistics relevant to reliability and convergent validity. As part of the measurement model evaluation, three (PEC4, CPLAY4 and RES4) of the 40 items were removed from the analysis due to low factor loadings (<0.40) (Hair et al., 2017). This improved the Average Variance Extracted (AVE) on the corresponding constructs from <0.60 to >0.70 in each case. To assess convergent validity, the AVE was examined. AVE represents the communality of a construct and should exceed 0.50 (Hair et al., 2017). AVE ranged between 0.686 and 0.897, suggesting convergent validity is well supported. Cronbach’s alpha and composite reliability measures all exceeded the benchmark of 0.70 (applicable to both measures), indicating the internal consistency of the study’s measures. Cronbach’s alpha scores ranged between 0.789 and 0.943, while composite reliability ranged between 0.896 and 0.958.

Discriminant analysis was first assessed by examining cross-loadings (not reported). This confirmed that each construct’s item loadings exceeded the corresponding cross-loadings for those items. Next, the Fornell-Larker criterion was applied. With this test, the square-root of each construct’s AVE is checked to see that it exceeds its highest correlation with any other construct. Table 3 presents the relevant statistics and confirms that the criterion is met for all constructs. Together, these results establish the discriminant validity of the study’s measures.

Table 2. Descriptive, reliability and validity statistics.

Construct	Item	Mean	S.D.	Range	Outer loading	Cronbach's alpha	Composite reliability	Average variance extracted
Perceived Usefulness (PU)	PU1	5.281	1.348	1 - 7	0.873	0.941	0.958	0.851
	PU2	5.193	1.249	1 - 7	0.942			
	PU3	5.070	1.449	1 - 7	0.942			
	PU4	5.281	1.496	1 - 7	0.930			
Perceived Ease of Use (PEOU)	PEOU1	5.228	1.545	1 - 7	0.865	0.900	0.929	0.767
	PEOU2	4.807	1.572	1 - 7	0.844			
	PEOU3	5.298	1.363	1 - 7	0.899			
	PEOU4	4.895	1.447	1 - 7	0.894			
Computer Self-Efficacy (CSE)	CSE1	5.930	1.255	2 - 7	0.929	0.878	0.923	0.799
	CSE2	6.018	1.318	2 - 7	0.897			
	CSE3	6.158	1.121	3 - 7	0.853			
Perceptions of External Control (PEC)	PEC1	4.982	1.192	2 - 7	0.698	0.789	0.876	0.705
	PEC2	5.263	1.222	1 - 7	0.898			
	PEC3	5.491	1.230	2 - 7	0.906			
Computer Playfulness (CPLAY)	CPLAY1	5.000	1.499	1 - 7	0.884	0.855	0.909	0.770
	CPLAY2	5.053	1.234	1 - 7	0.916			
	CPLAY3	4.982	1.277	2 - 7	0.830			
Computer Anxiety (CANX)	CANX1	2.421	1.555	1 - 7	0.882	0.887	0.897	0.686
	CANX2	2.719	1.755	1 - 7	0.792			
	CANX3	2.526	1.580	1 - 6	0.866			
	CANX4	2.386	1.519	1 - 7	0.766			
Subjective Norm (SN)	SN1	4.456	1.215	2 - 7	0.850	0.868	0.909	0.715
	SN2	4.614	1.196	1 - 7	0.803			
	SN3	5.211	1.294	1 - 7	0.896			
	SN4	5.316	1.273	1 - 7	0.831			
Image (IMG)	IMG1	4.860	1.146	2 - 7	0.823	0.846	0.906	0.763
	IMG2	4.877	1.077	3 - 7	0.883			
	IMG3	4.965	1.256	2 - 7	0.913			
Job Application Relevance (REL)	REL1	4.596	1.387	1 - 7	0.921	0.872	0.921	0.795
	REL2	4.982	1.177	1 - 7	0.880			
	REL3	4.421	1.350	1 - 7	0.872			
Result Demonstrability (RES)	RES1	5.246	1.274	2 - 7	0.861	0.826	0.896	0.743
	RES2	5.158	1.121	2 - 7	0.906			
	RES3	4.825	1.110	1 - 7	0.817			
Behavioural Intention (BI)	BI1	5.491	1.488	1 - 7	0.969	0.943	0.963	0.897
	BI2	5.614	1.424	1 - 7	0.953			
	BI3	5.491	1.428	1 - 7	0.919			

Structural model

In evaluating the structural (inner) model, the collinearity between each set of predictor constructs for each sub-part of the model was first examined. This revealed that all variance inflation factor (VIF) values were below the commonly accepted cut-off of 5.0. The highest VIF was between PU and BI (3.943). As noted earlier, bootstrapping was used to determine the significance of path coefficients. As shown in [Figure 2](#), six of the 13 paths

Table 3. Discriminant validity – Fornell-Larker criterion analysis.

	BI	CANX	CPLAY	CSE	IMG	PEC	PEOU	PU	REL	RES	SN
BI	0.947										
CANX	−0.244	0.828									
CPLAY	0.322	−0.435	0.878								
CSE	0.450	−0.624	0.642	0.894							
IMG	0.655	−0.039	0.240	0.371	0.874						
PEC	0.676	−0.364	0.515	0.677	0.590	0.840					
PEOU	0.453	−0.293	0.376	0.386	0.394	0.543	0.876				
PU	0.625	−0.284	0.342	0.442	0.490	0.630	0.829	0.922			
REL	0.726	−0.024	0.344	0.304	0.823	0.605	0.363	0.527	0.891		
RES	0.708	−0.253	0.261	0.482	0.733	0.689	0.480	0.614	0.661	0.862	
SN	0.753	−0.126	0.252	0.454	0.663	0.715	0.515	0.635	0.681	0.773	0.846

Note: The square roots of each construct's average variance extracted (AVE) is on the diagonal. The nondiagonal elements are correlations between constructs.

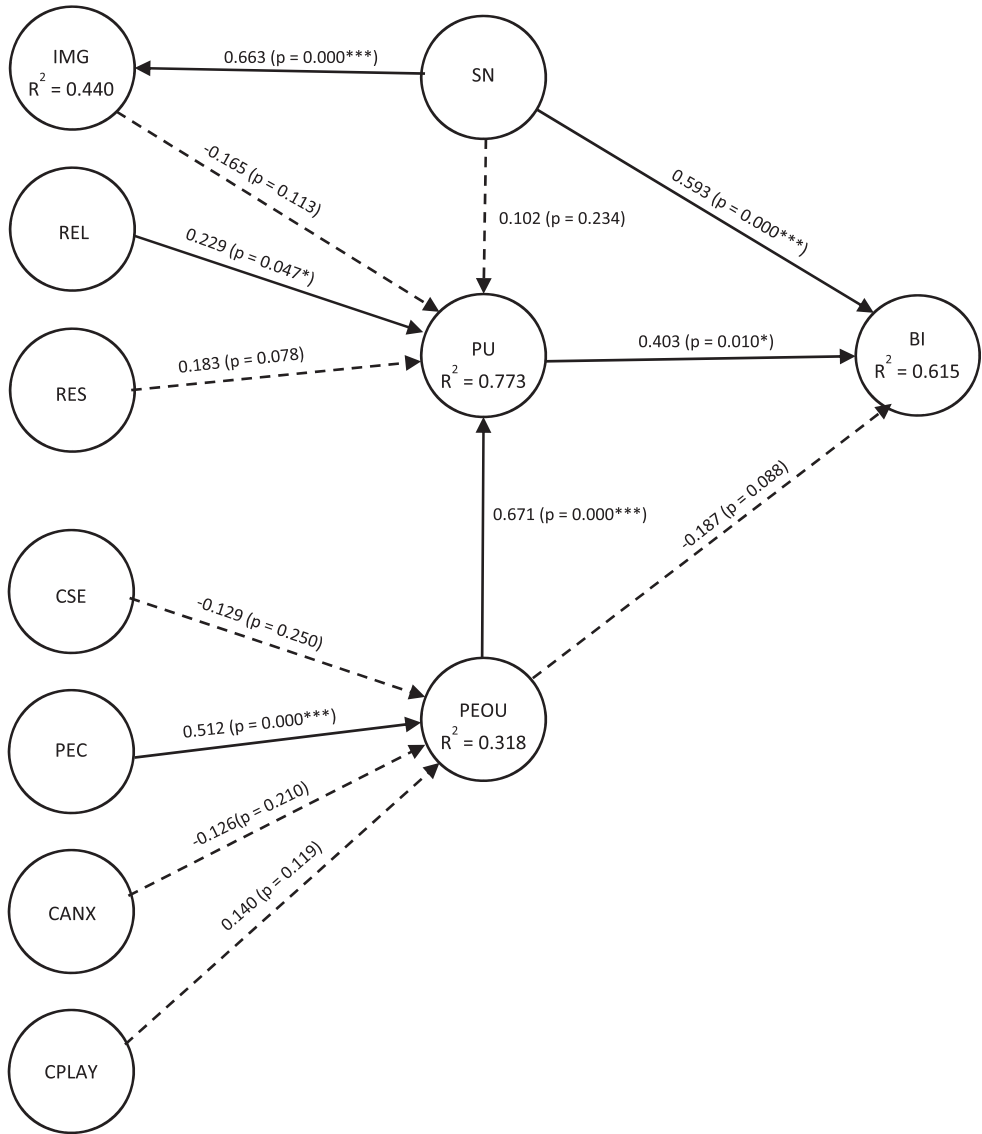
were significant at 5 percent level or lower. Additional analysis of bootstrap confidence intervals (not reported) confirmed the stability of the significant path coefficient estimates. Last, PLS aims to maximise the R^2 values of the endogenous latent variables in the path model (Hair et al., 2017). The R^2 values are shown within each construct's bubble in Figure 2. Based on Chin (1998) rules of thumb, the R^2 value for PU (0.773) may be considered *substantial*, the R^2 values for IMG (0.440) and BI (0.615) *moderate*, respectively, while the R^2 value for PEOU (0.318) *weak*.

Results of hypothesis testing and discussion

The results of hypothesis testing are presented in Table 4. Overall, six of the 13 hypotheses were supported. Unsurprisingly, perceived usefulness is a significant predictor of behavioural intention ($\beta = 0.403$, $p < 0.05$). Perceived ease of use, however, does not significantly influence students' intentions to use digital badges, although it does directly influence their perceptions of badge usefulness ($\beta = 0.671$, $p < 0.001$). These findings suggest that the perceived ease of use's effect on intention to use is indirect only, being mediated by perceived usefulness. The lack of a direct link between perceived ease of use and behavioural intention may have been a function of students' lack of hands-on experience with digital badges. Future studies might usefully examine the impact of digital badge experience on this relationship.

Table 4. Hypotheses testing results.

Structural relationship	Path coefficient	Standard deviation	T statistics	P values	Hypothesis	Hypothesis Supported?
CSE → PEOU	−0.129	0.191	0.674	0.250	H10	No
PEC → PEOU	0.512	0.136	3.772	0.000	H11	Yes
CANX → PEOU	−0.126	0.156	0.807	0.210	H12	No
CPLAY → PEOU	0.140	0.119	1.178	0.119	H13	No
SN → PU	0.102	0.140	0.725	0.234	H5	No
PEOU → PU	0.671	0.163	4.126	0.000	H3	Yes
IMG → PU	−0.165	0.136	1.21	0.113	H7	No
REL → PU	0.229	0.137	1.671	0.047	H8	Yes
RES → PU	0.183	0.129	1.419	0.078	H9	No
SN → IMG	0.663	0.072	9.217	0.000	H6	Yes
SN → BI	0.593	0.158	3.766	0.000	H4	Yes
PU → BI	0.403	0.175	2.309	0.010	H1	Yes
PEOU → BI	−0.187	0.138	1.356	0.088	H2	No



* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Figure 2. Caption: Results of the structural model. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

The standardised path coefficient for subjective norm of $\beta = 0.593$ ($p < 0.001$) indicates that it has a relatively greater effect than perceived usefulness on the behavioural intention of students to use digital badges as a means of showcasing their employability skills. Furthermore, subjective norm had no direct influence on perceived usefulness, and although it influenced image, image was not a determinant of perceived usefulness. Consequently, subjective norm appears to have no indirect effect on behavioural intention, only a direct effect. This suggests that, irrespective of their own perceptions about the usefulness of digital badges, perceptions of important others, including the university

itself, strongly influence students in determining their intention to use digital badges. That is, students are willing to subordinate their own views on the usefulness of digital badges for those of others. This ‘compliance’ orientation resonates with Venkatesh and Davis (2000, p. 188) who believe that this effect results ‘... whenever an individual perceives that a social actor wants him or her to perform a specific behaviour, and the social actor has the ability to reward the behaviour or punish nonbehaviour’. As digital badge use will in most cases be voluntary, it is not immediately obvious why students are willing to acquiesce to the perceived expectations of others. Voluntary use would be expected to increase if there is value in doing so, and that value is based on the importance placed by others on the display of badges. We speculate that beliefs about the university’s (hypothetical) initial and ongoing investment in badge technology, (including ongoing support) created a ‘social contract’ with students based on the norm of reciprocity (Gouldner, 1960). We encourage future researchers to confirm the precise mechanisms by which subjective norm influences intention to use digital badges.

As anticipated, job application relevance directly positively influenced perceived usefulness ($\beta = 0.229$, $p < 0.05$). However, result demonstrability failed to reach significance. The latter result may again reflect students’ lack of real-world experience with digital badges and suggests that when introducing digital badges, universities should present clear evidence (perhaps via employer and alumni endorsements) of their efficacy.

Focusing on determinants of perceived ease of use, only perceptions of environmental control had a significant influence. This result highlights the importance of universities living up to students’ expectations of the usefulness of adequate training and support services around the rollout and ongoing use of digital badges. Such resources can have a significant role in removing perceived barriers to technology use (Venkatesh et al., 2003).

Neither computer self-efficacy nor computer anxiety were significantly associated with perceived ease of use of digital badges. These results may reflect the student participants’ recent and relatively intense experience with computers. The survey was conducted towards the end of 2022, and due to the disruptive effects of COVID lockdowns throughout the entire period of their tertiary studies, all student participants had considerable exposure to many forms of online assessment and a range of e-learning tools. Moreover, due to the nature of the accounting programmes at their university, all students had had hands-on experience with a wide range of computer applications (spreadsheets, cloud accounting, data analytic applications, databases, etc.). Unsurprisingly, then, their average level of computer anxiety across all items was relatively low at 2.513 (on a 1–7 scale), while their mean level of computer self-efficacy across all items was relatively high at 6.035. In a recent study, Abdullah et al. (2016) found no effect of computer anxiety among student participants on the perceived ease of use of e-portfolio, which they attributed to the fact that surveyed students were all studying computer-related courses, in conjunction with the observation that the general increase in use of computer technology in every aspect of daily life was likely leading to a general decline in negative perceptions associated with computers. Shen and Eder (2009) make a similar observation.

Computer playfulness, an intrinsic motivation aspect of technology use, was also not a significant predictor of the perceived ease of use of digital badges. Hackbarth et al. (2003) argue that when users first interact with a technology, they typically feel somewhat intimidated and as a consequence, their degree of playfulness is low. However, as they gain

familiarity with the system, they are more likely to explore the technology and interact with it in a more spontaneous manner. The lack of hands-on experience with digital badges among students in our study may have mitigated the effect of playfulness. However, as badges become more prevalent, we might expect the effect of playfulness to increase. This merits further study.

Overall then, six of the hypothesised relationships were supported. As digital badges gain momentum and become more widely used, we might expect experience with them to moderate several of the relationships considered in this study. As already mentioned, it is expected to enhance the effects of playfulness over time, but also likely attenuate the direct effect of social norm on intention to use digital badges over time (Venkatesh & Bala, 2008). Consequently, we suggest that a longitudinal study might usefully examine these effects.

Venkatesh and Bala's (2008) study suggests a variety of interventions that can be undertaken by universities to influence many of the determinants of perceived usefulness and perceived ease of use of digital badges. The most relevant to the context of our study include having students involved in the design of badges, appropriate training and instruction in their use, and ongoing technical support – including the period after students have graduated. Indeed, the latter may be an important means through which universities remain connected with their alumni – an unintended but positive outcome.

Additional analyses

As has been previously discussed, the generic nature of extended versions of TAM, limits its scope in identifying barriers to use (Yousafzai et al., 2010). To this end, based on a review of the literature, a list of 13 issues specific to digital badges was presented to survey participants. These issues are listed in Table 5. Participants rated (on a 7-point Likert scale) the importance of each issue to their decision to use university-issued digital badges for employment purposes.

Table 5. Importance of specific issues to decision to use digital badges.

Issue	Mean	Standard Deviation	Range
1. The digital badges are issued to recognise specific meaningful achievements or accomplishments, based on approved standards and criteria.	5.661	1.299	1 - 7
2. The digital badges are accessible to employers and other external stakeholders.	5.714	1.235	2 - 7
3. The digital badges can be shared widely on social media sites, like LinkedIn.	5.625	1.383	1 - 7
4. The digital badges are able to be validated for authenticity.	6.161	1.082	3 - 7
5. The digital badges are quality assured and reflect high standards of integrity and credibility.	6.179	1.002	4 - 7
6. The digital badges are relevant to employers and other external stakeholders.	5.857	0.990	4 - 7
7. The digital badges are consistent in style.	5.089	1.392	1 - 7
8. The digital badges are comparable with those of other universities.	5.875	1.119	3 - 7
9. The digital badges are secure from manipulation.	6.071	1.116	4 - 7
10. The digital credential platform organisation (e.g. Credly, Accreditable, etc.) will not release personal information without my consent.	6.125	1.135	3 - 7
11. The digital credential platform organisation's future continuity.	5.857	1.156	3 - 7
12. The digital credential platform organisation is trustworthy.	6.125	1.151	2 - 7
13. The digital badges reliably differentiate achievements between students.	5.750	1.169	3 - 7

The importance of all items was assessed on a seven-point scale, ranging from 1 = Not at all important to 7 = Extremely important.

All thirteen items were on average rated above the scale midpoint of 4 (neither important nor unimportant). Of the five issues rated 6 (very important) or above, three reflected the somewhat related issues of perceived security risk (Issue 9), perceived privacy risk (Issue 10), and trustworthiness of third-party credential platforms with which many universities are likely to contract (Issue 12). These issues reflected similar concerns raised by both Coleman (2018) and Pitt et al. (2019). Prior research in other technology contexts suggest that both privacy and security risks contribute to trustworthiness, which in turn influences behavioural intention to use the focal technology (Ho et al., 2017; Wong & Mo, 2019; Yousafzai et al., 2010). Future research could examine the interplay and relative importance of these potentially critical factors in relation to digital badge use in higher education settings. The remaining two high importance issues (Issues 4 and 5) appear to relate to badge authenticatability concerns, and resonate with the concerns of Randall and West (2022), Coleman (2018), and others. The least important of the issues (Issue 7) related to an aesthetic concern, the consistency of badge style. Although it was the least important of the issues, its mean score of around 5 suggests that universities' design teams should not ignore this issue.

Conclusions

This exploratory study focused on the emergent use of digital badge technology in higher education. While digital badges may be deployed for a variety of purposes, the current study focused on their use as a tool for showcasing accounting student's attainment of employability skills. In general, accounting students intended to adopt digital badges to facilitate job application activities (the average item score for behavioural intention was 5.532 on a 7-point scale). The only two factors found to directly influence badge usage intentions were subjective norm and perceived usefulness, with the former having the greater relative effect. The perceived ease of use of digital badges only had an indirect effect on intention. The study also examined general determinants of perceived usefulness and perceived ease of use. The degree to which students could see the relevance of digital badges to the job application process positively influenced their perceptions of badge usefulness. Further, the extent to which they perceived that they had control over the use of their digital badges and that their university would well support them in using digital badges (i.e. perceptions of external control) positively influenced their belief that digital badges would be easy to use. Additional analyses examined the perceived importance of a range of specific issues to students. This revealed that students have strong beliefs about security and privacy issues associated with digital badges, in addition to a concern that badges be able to be authenticated. Overall the findings provide a basis for meaningful interventions by universities aimed at maximising the likelihood of badge use among their student bodies.

The study drew on the TAM3 model (Venkatesh & Bala, 2008). The findings suggest that the model developed in the study might usefully be extended to incorporate security, privacy, and trust concerns, particularly in relation to digital credential platform organisations whose infrastructure is likely to underpin many instances of digital badges in higher education.

While this study focused solely on students, the importance of employers to ultimate success of digital badges cannot be understated. Future research should consider

the issues from accounting employers' perspectives, including determining the value they place in digital badges and the factors that would facilitate their use in the recruitment process. Focus groups and interviews could be particularly insightful for this purpose.

This research is subject to the usual limitations of the survey method. Although great care was taken in design and analysis, data were obtained from a single university in New Zealand. Further, as student participation was voluntary, self-selection bias cannot be fully ruled out. More research is needed to extend the generalisability of the findings to other countries and discipline areas, and to survey a more comprehensive range of accounting students. Additionally, given the paucity of existing use cases of digital badges in higher education in general, and in accounting education in particular, few students, including those in this study, have had hands-on experience with digital badges. We expect that over time, experience with badges and their outcomes will change the relationships modelled in this study. Consequently, we encourage the use of longitudinal research designs. Finally, data for the study were obtained solely through the means of a survey questionnaire. Qualitative research approaches, such as those employing structured interviews and focus groups, may usefully complement the findings of this study.

Notes

1. Indeed, some critics have warned that the privileging of employability over citizenship “risks erosion of the higher purpose of universities and is to the detriment of students’ social and professional development” (Oliver & Jorre de St Jorre, 2018, p. 822). Notwithstanding this, employability skills remain a prominent influence on graduate attributes.
2. In their literature review of the employability skills literature, Curtis and McKenzie (2002) found that various alternatives for the descriptor, ‘skills’, have been used in the literature, including competencies, competences, attributes, characteristics and qualities. Similarly, a range of qualifiers have been used in place of ‘employability’, such as core, key, necessary, essential, generic, transferable, graduate, lifelong learning. For the purposes of this paper, we that ‘employability’ and ‘skills’ subsume these alternative qualifiers and descriptors, respectively.
3. Although *attitude* was included in the original TAM, it has subsequently been found to at best only partially mediate the relationships between both perceived ease of use and perceived usefulness and behavioural intention to use the focal technology (Davis et al., 1989). Subsequent versions of TAM have tended to omit the construct in favour of a more parsimonious model (see Abdullah et al., 2016, for further discussion). Consequently, the attitude construct is not included in the current model.
4. Although it is likely that those with greater familiarity with digital badges will have answered some the questions based on their personal experiences, we do not believe this will have biased the results. Only 1 (2%) participant rated themselves as ‘Very familiar’ with the technology, while 12 (21%) participants rated themselves as ‘Not at all familiar’. It is not uncommon for technology acceptance studies involving new or emerging technology to provide such information to participants (for example, see Aburbeian, Owda & Owda, 2022 and Jaschinski, Allouch, Peters, Cachucho & Van Dijk, 2021).
5. At the time of the research, the university did not yet use digital badges but offered an optional printed co-curricular record (CCR) to students. This is defined by the university as a statement of verified student engagement in university activities that are not part of an academic programme. The CCR maps the related ‘learning and development’ hours from CCR activities to various work readiness skills.
6. Approval to conduct this research was received from the University’s (at which the research was conducted) Human Research Ethics Committee, Approval number HREC 2022/54/LR-PS.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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References

- Abdullah, F., & Ward, R. (2016). Developing a general extended technology acceptance model for E-learning (GETAMEL) by analysing commonly used external factors. *Computers in Human Behavior*, 56, 238–256. <https://doi.org/10.1016/j.chb.2015.11.036>
- Abdullah, F., Ward, R., & Ahmed, E. (2016). Investigating the influence of the most commonly used external variables of TAM on students' Perceived Ease of Use (PEOU) and Perceived Usefulness (PU) of e-portfolios. *Computers in Human Behavior*, 63, 75–90. <https://doi.org/10.1016/j.chb.2016.05.014>
- Abebe, G., Caria, S. S., Fafchamps, M., Falco, P., Franklin, S., & Quinn, S. (2021). Anonymity or distance? Job search and labour market exclusion in a growing African city. *Review of Economic Studies*, 88(3), 1279–1310. <https://doi.org/10.1093/restud/rdaa057>
- Abramovich, S., Schunn, C., & Higashi, R. M. (2013). Are badges useful in education?: it depends upon the type of badge and expertise of learner. *Education Tech Research Dev*, 61(2), 217–232. <https://doi.org/10.1007/s11423-013-9289-2>
- Aburbeian, A. M., Owda, A. Y., & Owda, M. (2022). A technology acceptance model survey of the metaverse prospects. *AI*, 3(2), 285–302. <https://doi.org/10.3390/ai3020018>
- Accounting Education Change Commission. (1990). Objectives of education for accountants: Position statement number one. *Issues in Accounting Education*, 6(2), 307–312.
- Agnew, H. (2016, January 28). 'Big four' look beyond academics. *Financial Times*. <https://www.ft.com/content/b8c66e50-beda-11e5-9fdb-87b8d15baec2>
- Albrecht, W., & Sack, R. (2000). *Accounting education: Charting the course through a perilous future*. American Accounting Association.
- American Accounting Association. (1986). Committee on the future structure, content and scope of accounting education [The bedford committee]. *Future accounting education: Preparing for the expanding profession*. *Issues in Accounting Education*, 1(1), 168–195.
- Barclay, D., Higgins, C., & Thompson, R. (1995). The Partial Least Squares (PLS) approach to causal modeling: Personal computer adoption and use as an illustration. *Technology Studies*, 2(2), 285–309.
- Bassi, V., & Nansamba, A. (2022). Screening and signalling non-cognitive skills: Experimental evidence from Uganda. *The Economic Journal*, 132(642), 471–511. <https://doi.org/10.1093/ej/ueab071>
- Birkett, W. (1993). *Competency based standards for professional accountants in Australia and New Zealand*. ACPA and ICAA.
- Bowden, J., Hart, G., King, B., Trigwell, K., & Watts, O. (2000). *Generic capabilities of ATN university graduates*. Canberra: Australian Government Department of Education, Training and Youth Affairs. <http://www.clt.uts.edu.au/atn.grad.cap.project.index.html>
- Bowen, K., & Thomas, A. (2014). Badges: a common currency for learning. *Change: The magazine of higher learning*, 46(1), 21–25.
- Bui, B., & Porter, B. (2010). The expectation-performance gap in accounting education: An exploratory study. *Accounting Education*, 19(1-2), 23–50. <https://doi.org/10.1080/09639280902875556>
- Carey, K. L., & Stefaniak, J. E. (2018). An exploration of the utility of digital badging in higher education settings. *Educational Technology Research and Development*, 66(5), 1211–1229. <https://doi.org/10.1007/s11423-018-9602-1>

- Carranza, E., Garlick, R., Orkin, . K., & Rankin, N. (2022). Job search and. hiring with limited information about workseekers' skills. *American Economic Review*, 112(11), 3547–3583. <https://doi.org/10.1257/aer.20200961>.
- Castrillon, C. (2021). How to use social media to job search. *Forbes*. <https://www.forbes.com/sites/carolinecastrillon/2021/03/24/how-to-use-social-media-to-job-search/?sh=733072a43cfd>
- Chaplin, S. (2017). Accounting education and the prerequisite skills of accounting graduates: Are accounting firms' moving the boundaries? *Australian Accounting Review*, 27(1), 61–70. <https://doi.org/10.1111/auar.12146>
- Chen, M. Y., Chang, F. M. T., Chen, C. C., Huang, M. J., & Chen, J. W. (2012). Why do individuals use e-Portfolios. *Educational Technology & Society*, 15(4), 114–125.
- Chen, Z. S. C., Yang, S. J. H., & Huang, J. J. S. (2015). Constructing an e-portfolio-based integrated learning environment supported by library resource. *The Electronic Library*, 33(2), 273–291. <https://doi.org/10.1108/EL-07-2013-0118>
- Chin, W. W. (1998). *The Partial Least Squares approach to structural equation modeling*. Lawrence Erlbaum.
- Chin, W. W., Marcolin, B., & Newsted, P. (2003). A Partial Least Squares latent variable modeling approach for measuring interaction effects: Results from a Monte Carlo Simulation study and an electronic-mail emotion/adoption study. *Information Systems Research*, 14(2), 189–217. <https://doi.org/10.1287/isre.14.2.189.16018>
- Cohen, J. (1992). A power primer. *Psychological Bulletin*, 112(1), 155–159. <https://doi.org/10.1037/0033-2909.112.1.155>
- Coleman, J. D. (2018). Engaging undergraduate students in a co-curricular digital badging platform. *Education and Information Technologies*, 23(1), 211–224. <https://doi.org/10.1007/s10639-017-9595-0>
- Craig, R., & Amernic, J. (2002). Accountability of accounting educators and the rhythm of the university: Resistance strategies for postmodern blues. *Accounting Education*, 11(2), 121–171. <https://doi.org/10.1080/0963928021000031772>
- Curtis, D., & McKenzie, P. (2002). *Employability skills for Australian industry: Literature review and framework development. Report to Business Council of Australia; Australian Chamber of Commerce and Industry*. Science and Training.
- Davis, F., Bagozzi, R. P., & Warshaw, P. R. (1989). User acceptance of computer technology: A comparison of two theoretical models. *Management Science*, 35(8), 982–1003. <https://doi.org/10.1287/mnsc.35.8.982>
- Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 319–340.
- Dyjur, P., & Lindstrom, G. (2017). Perceptions and uses of digital badges for professional learning development in higher education. *TechTrends*, 61(4), 386–392. <https://doi.org/10.1007/s11528-017-0168-2>
- European Commission. (2021). *A European approach to micro-credentials*. European Commission. <https://education.ec.europa.eu/sites/default/files/2022-01/micro-credentials%20brochure%20updated.pdf>.
- Gamrat, C., Zimmerman, H. T., Dudek, J., & Peck, K. (2014). Personalized workplace learning: An exploratory study on digital badging within a teacher professional development program. *British Journal of Educational Technology*, 45(6), 1136–1148. <https://doi.org/10.1111/bjet.12200>
- Gefen, D., Rigdon, E., & Straub, D. (2011). Editor's comments: An update and extension to SEM guidelines for administrative and social science research. *MIS Quarterly*, 35(2), iii. <https://doi.org/10.2307/23044042>
- Gerber, N., Gerber, P., & Volkamer, M. (2018). Explaining the privacy paradox: A systematic review of literature investigating privacy attitude and behavior. *Computers & Security*, 77, 226–261. <https://doi.org/10.1016/j.cose.2018.04.002>
- Gibson, D., Ostashewski, N., Flintoff, K., Grant, S., & Knight, E. (2015). Digital badges in Education. *Education and Information Technologies*, 20(2), 403–410. <https://doi.org/10.1007/s10639-013-9291-7>

- Gogel, D., Eikenaar, J. H., & Rajabi, P. (2020). Badging for accreditation: Electronic credentialing in the undergraduate curriculum. *Proceedings of the Canadian Engineering Education Association (CEEAA)*.
- Goligoski, E. (2012). Motivating the learner: Mozilla's open badges program. *Access to Knowledge: A Course Journal*, 4.
- Goodhue, D., Lewis, W., & Thompson, R. (2012). Comparing PLS to regression and LISREL: A response to marcoulides, chin, and saunders. *MIS Quarterly*, 36(3), 703–716. <https://doi.org/10.2307/41703476>
- Gouldner, A. W. (1960). The norm of reciprocity: a preliminary statement. *American Sociological Review*, 25(2), 161–178.
- Granić, A., & Marangunić, N. (2019). Technology acceptance model in educational context: A systematic literature review. *British Journal of Educational Technology*, 50(5), 2572–2593. <https://doi.org/10.1111/bjet.12864>
- Hackbarth, G., Grover, V., & Mun, Y. Y. (2003). Computer playfulness and anxiety: Positive and negative mediators of the system experience effect on perceived ease of use. *Information & Management*, 40(3), 221–232. [https://doi.org/10.1016/S0378-7206\(02\)00006-X](https://doi.org/10.1016/S0378-7206(02)00006-X)
- Hair, J. J. F., Hult, G. T. M., Ringle, C. M., & Sarstedt, M. (2017). *A primer on partial least squares structural equation modeling (PLS-SEM)*. Sage Publications.
- Hartnett, M. (2021). How and why are digital badges being used in higher education in New Zealand? *Australasian Journal of Educational Technology*, 37(3), 104–118. <https://doi.org/10.14742/ajet.6098>
- Hayes, S., Freudenberg, B., & Delaney, D. (2022). Work ready graduates for Australian small and medium accounting firms. *Journal of Teaching and Learning for Graduate Employability*, 13(1), 1–19. <https://doi.org/10.21153/jtlge2022vol13no1art1026>
- Hill, J., & Walkington, H. (2016). Developing graduate attributes through participation in undergraduate research conferences. *Journal of Geography in Higher Education*, 40(2), 222–237. <https://doi.org/10.1080/03098265.2016.1140128>
- Ho, S. M., Ocasio-Velázquez, M., & Booth, C. (2017). Trust or consequences? Causal effects of perceived risk and subjective norms on cloud technology adoption. *Computers & Security*, 70, 581–595. <https://doi.org/10.1016/j.cose.2017.08.004>
- Hope, J. (2016). Support campuswide educational goals with transcript enhancements. *The Successful Registrar*, 16(7), 1–5.
- Horst, M., Kuttschreuter, M., & Gutteling, J. M. (2007). Perceived usefulness, personal experiences, risk perception and trust as determinants of adoption of e-government services in The Netherlands. *Computers in Human Behavior*, 23(4), 1838–1852. <https://doi.org/10.1016/j.chb.2005.11.003>
- Jackson, D., Michelson, G., Munir, R., & Richardson, J. (2021). *Sustainable graduate accounting and finance career pathways: A multi-stakeholder approach*. CPA.
- Jaschinski, C., Allouch, S. B., Peters, O., Cachucho, R., & van Dijk J. A. G. M. (2021). Acceptance of technologies for aging in place: A conceptual model. *Journal of Medical Internet Research*, 23(3): e22613. <https://doi.org/10.2196/22613>. PMID: 33787505; PMCID: PMC8047804
- Jorgensen, N. (2023). The power of social media in recruiting. *Forbes*. <https://www.forbes.com/sites/forbeshumanresourcescouncil/2023/01/12/the-power-of-social-media-in-recruiting/?sh=268dd6762e92>.
- Jorre de St Jorre, T. (2020). Sharing achievement through digital credentials: Are universities ready for the transparency afforded by a digital world? *Re-imagining University Assessment in a Digital World*, 277–288.
- Jovanovic, B. (1979). Job matching and the theory of turnover. *Journal of Political Economy*, 87(5), 972–990. <https://doi.org/10.1086/260808>
- Karahanna, E., & Straub, D. W. (1999). The psychological origins of perceived usefulness and ease-of-use. *Information & Management*, 35(4), 237–250. [https://doi.org/10.1016/S0378-7206\(98\)00096-2](https://doi.org/10.1016/S0378-7206(98)00096-2)
- Kavanagh, M., & Drennan, L. (2008). What skills and attributes does an accounting graduate need? Evidence from student perceptions and employer expectations. *Accounting & Finance*, 48(2), 279–300. <https://doi.org/10.1111/j.1467-629X.2007.00245.x>

- Kensington-Miller, B., Knewstubb, B., Longley, A., & Gilbert, A. (2018). From invisible to SEEN: A conceptual framework for identifying, developing and evidencing unassessed graduate attributes. *Higher Education Research & Development*, 37(7), 1439–1453. <https://doi.org/10.1080/07294360.2018.1483903>
- Low, M., Botes, V., Rue, D. D., & Allen, J. (2016). Accounting employers' expectations-the ideal accounting graduates. *e-Journal of Business Education and Scholarship of Teaching*, 10(1), 36–57.
- Mahon, D. (2022). The role of graduate attributes in higher education. A review of the issues associated with graduate attributes and the case for their measurement. *Interchange*, 53(3-4), 353–369. <https://doi.org/10.1007/s10780-022-09457-5>
- Maina, M., Guàrdia, L., & Fernández-Ferrer, M. (2021). *Integrating employability in higher education*. EPICA. https://openaccess.uoc.edu/bitstream/10609/124366/12/integrating_employability.pdf.
- Menz, M. (2020). Integrating academic skills and employability – Revisiting the learning journal. *Journal of Perspectives in Applied Academic Practice*, 8(1), 115–120. <https://doi.org/10.14297/jpaap.v8i1.393>
- Miller, K. K., Jorre de St Jorre, T., West, J. M., & Johnson, E. D. (2020). The potential of digital credentials to engage students with capabilities of importance to scholars and citizens. *Active Learning in Higher Education*, 21(1), 11–22. <https://doi.org/10.1177/1469787417742021>
- Moore, G. C., & Benbasat, I. (1991). Development of an instrument to measure the perceptions of adopting an information technology innovation. *Information Systems Research*, 2(3), 192–222. <https://doi.org/10.1287/isre.2.3.192>
- Oliver, B. (2011). *Assuring graduate outcomes good practice*. Australian Learning and Teaching Council. https://www.jcu.edu.au/__data/assets/pdf_file/0019/122842/jcu_086521.pdf.
- Oliver, B., & Jorre de St Jorre, T. (2018). Graduate attributes for 2020 and beyond: Recommendations for Australian higher education providers. *Higher Education Research & Development*, 37(4), 821–836. <https://doi.org/10.1080/07294360.2018.1446415>
- Overtom, C. (2000). Employability skills: An update. Center on Education and Training for Employment. ERIC Digest, No. 220. <https://files.eric.ed.gov/fulltext/ED445236.pdf>.
- Padilla-Meléndez, A., del Aguila-Obra, A. R., & Garrido-Moreno, A. (2013). Perceived playfulness, gender differences and technology acceptance model in a blended learning scenario. *Computers & Education*, 63, 306–317. <https://doi.org/10.1016/j.compedu.2012.12.014>
- Parker, H. E. (2015). Digital badges to assess bloom's affective domain. *The National Teaching & Learning Forum*, 24(4), 9–11. <https://doi.org/10.1002/ntlf.30031>
- Pitt, C., Bell, A., Onofre, E., & Davis, K. (2019). A badge, not a barrier. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. Paper No., 690, 1–14. <https://doi.org/10.1145/3290605.3300920>.
- Randall, D., & West, R. (2022). Who cares about open badges? An examination of principals' perceptions of the usefulness of teacher open badges in the United States. *Open Learning: The Journal of Open, Distance and e-Learning*, 37(1), 65–83. <https://doi.org/10.1080/02680513.2020.1752166>
- Schepers, J., & Wetzels, M. (2007). A meta-analysis of the technology acceptance model: Investigating subjective norm and moderation effects. *Information & Management*, 44(1), 90–103. <https://doi.org/10.1016/j.im.2006.10.007>
- Shen, J., & Eder, L. B. (2009). Intentions to use virtual worlds for education. *Journal of Information Systems Education*, 20(20), 225.
- Shields, R., & Chugh, R. (2017). Digital badges – Rewards for learning? *Education and Information Technologies*, 22(4), 1817–1824. <https://doi.org/10.1007/s10639-016-9521-x>
- Sin, S., & McGuigan, N. (2013). Fit for purpose: A framework for developing and assessing complex graduate attributes in a changing higher education environment. *Accounting Education*, 22(6), 522–543. <https://doi.org/10.1080/09639284.2013.847320>
- Singhal, P. (2017). PwC to end university degree employment requirement. <https://www.smh.com.au/education/pwc-to-end-university-degree-employment-requirement-20170424-gvrb7c.html>.

- Sousa-Vieira, M., Ferrero-Castro, D., & López-Ardao, J. (2022). Design, development and Use of a digital badges system in higher education. *Applied Sciences*, 12(1), 220–236. <https://doi.org/10.3390/app12010220>
- Star, C., & Hammer, S. (2008). Teaching generic skills: Eroding the higher purpose of universities, or an opportunity for renewal? *Oxford Review of Education*, 34(2), 237–251. <https://doi.org/10.1080/03054980701672232>
- Sumak, B., Hericko, M., & Pusnik, M. (2011). A meta-analysis of e-learning technology acceptance: The role of user types and e-learning technology types. *Computers in Human Behavior*, 27(6), 2067–2077. <https://doi.org/10.1016/j.chb.2011.08.005>
- Sykes, T. A., Venkatesh, V., & Johnson, J. L. (2014). Enterprise system implementation and employee job performance: Understanding the role of advice networks. *MIS Quarterly*, 38(1), 51–72. <https://doi.org/10.25300/MISQ/2014/38.1.03>
- Tan, L. M., Laswad, F., & Chua, F. (2022). Bridging the employability skills gap: Going beyond classroom walls. *Pacific Accounting Review*, 34(2), 225–248. <https://doi.org/10.1108/PAR-04-2021-0050>
- Taylor, S., & Todd, P. A. (1995). Understanding information technology usage: A test of competing models. *Information Systems Research*, 6(2), 144–176. <https://doi.org/10.1287/isre.6.2.144>
- Tempone, I., Kavanagh, M., Segal, N., Hancock, P., Howieson, B., & Kent, J. (2012). Desirable generic attributes for accounting graduates into the twenty-first century. *Accounting Research Journal*, 25(1), 41–55. <https://doi.org/10.1108/10309611211244519>
- Teo, T. (2010). Examining the influence of subjective norm and facilitating conditions on the intention to use technology among pre-service teachers: A structural equation modeling of an extended technology acceptance model. *Asia Pacific Education Review*, 11(2), 253–262. <https://doi.org/10.1007/s12564-009-9066-4>
- Tomić, B., Jovanović, J., Milikić, N., Devedžić, V., Dimitrijević, S., Đurić, D., & Ševarec, Z. (2017). Grading students' programming and soft skills with open badges: A case study. *British Journal of Educational Technology*, 50(2), 518–530.
- Unal, E., & Uzun, A. M. (2021). Understanding university students' behavioral intention to use Edmodo through the lens of an extended technology acceptance model. *British Journal of Educational Technology*, 52(2), 619–637. <https://doi.org/10.1111/bjet.13046>
- Ursavaş, ÖF, Yalçın, Y., & Bakır, E. (2019). The effect of subjective norms on preservice and in-service teachers' behavioural intentions to use technology: A multigroup multimodel study. *British Journal of Educational Technology*, 50(5), 2501–2519. <https://doi.org/10.1111/bjet.12834>
- Venkatesh, V. (2000). Determinants of perceived ease of use: Integrating control, intrinsic motivation, and emotion into the technology acceptance model. *Information Systems Research*, 11(4), 342–365. <https://doi.org/10.1287/isre.11.4.342.11872>
- Venkatesh, V., & Bala, H. (2008). Technology acceptance model 3 and a research agenda on interventions. *Decision Sciences*, 39(2), 273–315. <https://doi.org/10.1111/j.1540-5915.2008.00192.x>
- Venkatesh, V., & Davis, F. D. (2000). A theoretical extension of the technology acceptance model: Four longitudinal field studies. *Management Science*, 46(2), 186–204. <https://doi.org/10.1287/mnsc.46.2.186.11926>
- Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly*, 27(3), 425–478. <https://doi.org/10.2307/30036540>
- Wilson, B. G., Gasell, C., Ozyer, A., & Scrogan, L. (2016). Adopting digital badges in higher education: Scoping the territory. In N. B.-M. D. Ifenthaler, & D.-K. Mah (Eds.), *Foundation of digital badges and micro-credentials* (pp. 163–177). Springer.
- Wong, W. H., & Mo, W. Y. (2019). A study of consumer intention of mobile payment in Hong Kong, based on perceived risk, perceived trust, perceived security and Technological Acceptance Model. *Journal of Advanced Management Science*, 7(2), 33–38. <https://doi.org/10.18178/joams.7.2.33-38>
- Yousafzai, S. Y., Foxall, G. R., & Pallister, J. G. (2010). Explaining internet banking behavior: Theory of reasoned action, theory of planned behavior, or technology acceptance model? *Journal of Applied Social Psychology*, 40(5), 1172–1202. doi:10.1111/j.1559-1816.2010.00615.x

Appendix

1. Construct items.

Construct		Item	Adapted from
Perceived Usefulness	PU1	Using university-issued digital badges would increase my employability when I apply for jobs.	(Davis, 1989; Venkatesh & Bala, 2008)
	PU2	Using university-issued digital badges would increase the quality of my job applications.	
	PU3	Using university-issued digital badges would enhance my effectiveness when I apply for jobs.	
	PU4	I would find using university-issued digital badges useful when applying for jobs.	
Perceived Ease of Use	PEOU1	If I were to obtain university-issued digital badges, my interaction with them would be clear and understandable.	(Davis, 1989; Venkatesh & Bala, 2008)
	PEOU2	If I were to obtain university-issued digital badges, interacting with them would not require a lot of my mental effort.	
	PEOU3	If I were to obtain university-issued digital badges, I would find them easy to use.	
	PEOU4	If I were to obtain university-issued digital badges, I would find it easy to get them to do what I want them to do.	
Computer Self-Efficacy	CSE1	I could complete a job or task using a computer if there was no one around to tell me what to do.	(Sykes et al., 2014)
	CSE2	I could complete a job or task using a computer if I could contact someone if I got stuck.	
	CSE3	I could complete a job or task using a computer if I had a lot of time to complete the job for which the software was provided.	
Perceptions of External Control	PEC1	I would have control over my university-issued digital badges.	(Venkatesh & Bala, 2008)
	PEC2	I would have the resources necessary to use university-issued digital badges.	
	PEC3	Given the resources, opportunities and knowledge it takes to use university-issued digital badges, it would be easy for me to use university-issued digital badges.	
	PEC4	University-issued digital badges are not compatible with other means I use for communicating my skills and competencies to potential employers.	
Computer Playfulness	CPLAY1	I would characterise myself as spontaneous when using computers.	(Venkatesh & Bala, 2008)
	CPLAY2	I would characterise myself as creative when using computers.	
	CPLAY3	I would characterise myself as playful when using computers.	
	CPLAY4	I would characterise myself as unoriginal when using computers.	
Computer Anxiety	CANX1	Computers do not scare me at all.	(Venkatesh & Bala, 2008)
	CANX2	Working with a computer makes me nervous.	
	CANX3	Computers make me feel uncomfortable.	
	CANX4	Computers make me feel uneasy.	
Subjective Norm	SN1	People who influence my behaviour think that I should use university-issued digital badges when applying for jobs.	(Venkatesh & Bala, 2008; Venkatesh et al., 2003)
	SN2	People who are important to me think that I should use university-issued digital badges when applying for jobs.	

(Continued)

Continued.

Construct		Item	Adapted from
Image	SN3	My university would be helpful in using university-issued digital badges for job applications.	(Venkatesh & Bala, 2008)
	SN4	In general, my university would support the use of university-issued digital badges for job applications. Employers think that I should use university-issued digital badges when applying for jobs.	
	IMG1	Students who use university-issued digital badges when applying for jobs are likely to have more prestige than those who do not.	
	IMG2	Students who use university-issued digital badges when applying for jobs will have a high profile.	
	IMG3	Being able to display university-issued digital badges when applying for jobs would be a status symbol for students.	
Job Application Relevance	REL1	When job searching, usage of university-issued digital badges is important.	(Venkatesh & Bala, 2008)
	REL2	When job searching, usage of university-issued digital badges is relevant.	
	REL3	The use of university-issued digital badges is pertinent to my job searching activities.	
Result Demonstrability	RES1	I would have no difficulty telling others about the results of using university-issued digital badges.	(Venkatesh & Bala, 2008)
	RES2	I believe I could communicate to others the consequences for me of using university-issued digital badges.	
	RES3	The results of using university-issued digital badges would be apparent to me.	
	RES4	I would have difficulty explaining why using university-issued digital badges for job searching may or may not be beneficial.	
Behavioral Intention	BI1	Assuming I had access to university-issued digital badges I intend to use them when I apply for jobs.	(Venkatesh & Bala, 2008; Venkatesh & Davis, 2000)
	BI2	Given that I had access to university-issued digital badges, I predict that I would use them when I apply for jobs.	
	BI3	If I had access to university-issued digital badges, I would plan to use the when I apply for jobs in the next 12 months.	

Note: All items were measured on a 7-point Likert scale where 1 = strongly disagree; 2 = moderately disagree, 3 = somewhat disagree, 4 = neutral (neither disagree nor agree), 5 = somewhat agree, 6 = moderately agree, and 7 = strongly agree.